

Article

Wind Performance of New and Existing Continuous Beam Bridges During Construction Stages

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Abstract: This study assesses the wind resistance and vortex-induced vibration (VIV) risks of the Dongzhou River Bridge in China reconstruction during critical construction stages. Computational Fluid Dynamics (CFD) simulations analyzed wind effects when the twin main girders were maximally separated, revealing asymmetric vortex shedding patterns influenced by upstream–downstream aerodynamic interactions. The upstream girder's wake generated complex flow fields, increasing turbulence on the downstream girder and indicating elevated VIV susceptibility. A 1:50 scale aeroelastic model validated these findings through wind tunnel tests, confirming that CFD-predicted critical VIV wind speeds aligned with experimental observations. Tests identified a distinct “jump-like” vibration mode at specific wind speeds (35–40 m/s full-scale equivalent), characterized by abrupt amplitude escalation rather than gradual growth—a signature of unstable VIV resonance. However, measured amplitudes remained below the 61.5 mm full-scale equivalent safety threshold, confirming that vibrations posed no critical risk. While aerodynamic coupling between girders requires monitoring during cantilever construction, the study concludes that existing control measures ensure safe construction and operation without structural modifications. These results provide actionable guidelines for wind risk mitigation through construction sequencing and real-time wind speed restrictions.

Keywords: wind tunnel tests; numerical simulation; full bridge aeroelastic model; new and existing bridge construction; continuous box girder bridge



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1. Introduction

With the improvement in people's living standards, the widespread use of automobiles has significantly increased, which has led to the inability of some existing bridges to meet the current traffic demands. Consequently, common solutions include widening the existing bridges or constructing new bridges at the upstream or downstream of the old bridge locations to accommodate larger traffic volumes. However, a common issue arises: the construction sequence between the old and new bridges and the inevitable gap between them may cause aerodynamic interference between the upstream and downstream bridges. Compared to single-span bridges, this creates more complex flow field characteristics and adverse factors that can trigger vortex-induced vibrations in the bridges [1]. Therefore, the aerodynamic interference effects between multiple-span bridges have attracted significant attention from researchers. Effective strategies to avoid mutual interference between bridges and optimize the construction sequence to ensure safety during the construction process have become crucial.

Zdravkovich et al. [2,3] conducted experimental investigations on flow interference between two circular cylinders, revealing the existence of a critical spacing that significantly

alters the vortex shedding characteristics of the upstream cylinder. The study further identified discontinuous “jumps” in flow parameters at specific critical spacing configurations. Watzinger et al. [4] studied the aerodynamic interference effects of two cylinders through both experiments and numerical simulations, revealing that both the upstream and downstream cylinders experienced vibration under the same experimental conditions. Du Xiaoqing et al. [5] found, through wind tunnel tests on closely spaced parallel square prisms, that the aerodynamic force time history of wide trailing-edge prisms is stable, and the intensity of vortex shedding is weak. The inner wind pressure distribution is highly sensitive to the distance ratio. Kiya et al. [6] suggested that the Strouhal number for vortex shedding could be calculated within a certain distance range between two cylinders. Yang Qun et al. [7] and others conducted wind tunnel pressure tests on serially arranged two and three square prisms at various distance ratios. Their research found that the flow characteristics of two and three prisms are completely different, and each has its own critical distance ratio. When the downstream prism spacing is smaller than the critical distance ratio, the drag coefficient, lift coefficient, and Strouhal number all decrease, while the wind pressure coefficient increases. When the spacing exceeds the critical distance ratio, the opposite trend is observed.

Currently, research on aerodynamic interference effects for separated main girder bridges has primarily focused on dual-girder cross-sections. However, during the construction of new and old bridges, a common scenario involves parallel operation of dual-girder bridges. Liu et al. [8,9], through wind tunnel experiments and numerical simulations, found that aerodynamic interference effects not only impact static wind loads but also significantly influence VIV. Argentini et al. [10], in sectional wind tunnel studies of the Ewijk Bridge, observed that, during construction, the flow field between parallel bridges was complex and unpredictable, leading to significant changes in the VIV of upstream box girders and the forces acting on downstream bridge segments. Seo et al. [11] identified and successfully reproduced a type of VIV observed on a dual-girder cable-stayed bridge using wind tunnel experiments. Alternating vortex shedding was found to amplify vibrations of the upstream main girder. Similarly, Park et al. [12] demonstrated, through wind tunnel experiments, that motion-induced VIV in downstream girders could amplify the VIV amplitude of upstream girders. Zhu et al. [13,14], via wind tunnel tests, revealed that the VIV lock-in range and maximum amplitude of dual-girder bridges increased under aerodynamic interference, with downstream girders experiencing significantly greater interference than upstream ones. In the experiment, it was observed that the vertical vibration of the main beam was more easily separated, forming a vortex. The amplitude of the downstream bridge is amplified when there are parallel Bridges [10,15,16]. Lin et al. [17,18], using CFD numerical simulations, studied dual-girder bridges under varying wind attack angles, showing that drag and lift coefficients were highly sensitive to aerodynamic interference effects. Honda et al. [19] identified an amplification effect of aerodynamic interference on VIV between three parallel bridges in wind tunnel experiments. Qin et al. [20], through wind tunnel studies, observed significant differences in the self-excited vibrations of separated box girders at varying spacings. Kwok et al. [21] found that changes in wind attack angles had a pronounced influence on the self-excited vibrations of separated box girders through pressure measurement experiments. Li et al. [22,23], based on field measurements of the Xihoumen Bridge, identified that vortices primarily formed at the central slot and the lower surface of the downstream girder during VIV events. The bridge should also be monitored during the construction process to ensure its reliability during construction [24].

Zheng et al. [25] and Ma et al. [26] used numerical computation methods to study the vortex-induced vibration performance of a 5:1 rectangular cross-section. Duc et al. [27] explored Wind tunnel tests and CFD technology were used to study the flow field around

the bridge deck with and without wing plates, and the results were compared. Li et al. [28] took a separated twin-box girder bridge as the research object and used segmental model vibration and pressure measurement tests to obtain surface wind pressures under different states of the separated twin-box girder. Kim et al. [29] conducted an in-depth study on the potential interference vortex-induced vibrations that may occur between two parallel cable-stayed bridges. Laima et al. [30] conducted sectional model experiments to study the effect of the spacing between separated double-box girders.

In summary, current studies on the aerodynamic interference effects of multi-girder bridges are primarily conducted through wind tunnel experiments and numerical simulations. Research findings indicate that aerodynamic interference significantly alters the VIV characteristics of parallel multi-girder bridges compared to single-girder bridges, potentially compromising their safety performance. Moreover, existing design codes for wind resistance are predominantly tailored to single-girder bridges, with no specific considerations for multi-girder configurations. Thus, research on aerodynamic interference effects of multi-girder bridges holds substantial engineering significance. It can provide critical insights during the initial design phase of new and old bridges, enabling better control of wind resistance performance and ensuring higher-quality wind resistance for bridges.

This study adopts an integrated CFD-experimental methodology to analyze the aerodynamic behavior of bridges during construction phases. The investigation implements a sequential validation framework: CFD simulations establish baseline aerodynamic coefficients, followed by wind tunnel testing for empirical verification and quantitative refinement. Ultimately, construction-phase safety recommendations are derived from this dual approach, providing designers with aerodynamic stabilization strategies to ensure uninterrupted construction operations.

2. Engineering Background

The Dongzhou River Bridge, with a total length of 415 m and a deck width of 32 m, consists of a superstructure comprising one prestressed concrete simply supported T-beam in the first span and six spans of frame-type prestressed T-beams with cantilevered beams. The cantilevered beams are also prestressed concrete T-beams. The substructure includes frame-type four-column square piers, integral pile caps, and driven pile foundations. Due to the existing span arrangement meeting the requirements of a Class III navigation clearance (The channel width is not less than 50 m) but failing to comply with the future planning requirements for a Class II navigation clearance (The channel width is not less than 100 m), the old bridge needs to be demolished and reconstructed. The old bridge is shown in Figure 1.

The span arrangement of the reconstructed and expanded New Dongzhou River Bridge is designed as a continuous steel–concrete composite girder with spans of 58 m + 63.5 m + 2 × 115 m + 63.5 m. The recommended structural form of the main girder is a variable-depth single-box single-cell steel box girder with straight webs. The left carriageway consists of three single-box girders, with a total width of 26 m, while the right carriageway consists of two single-box girders, with a single-girder width of 28.5 m. The main girder height is 2.8 m at mid-span and increases to 4.8 m at the pier locations. The substructure employs reinforced concrete (RC) Y-shaped thin-walled piers. The foundation system comprises an integral rectangular pile cap connected to driven steel pipe piles with a diameter of 2 m. Figure 2 illustrates the cross-section of the old bridge, and Figure 3 presents the cross-section of the new bridge.



Figure 1. The Old Dongzhou River bridge.

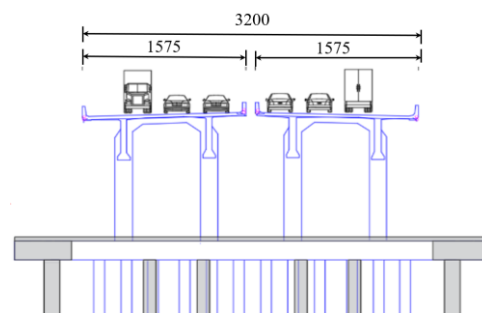


Figure 2. Cross-section of the old bridge (cm).

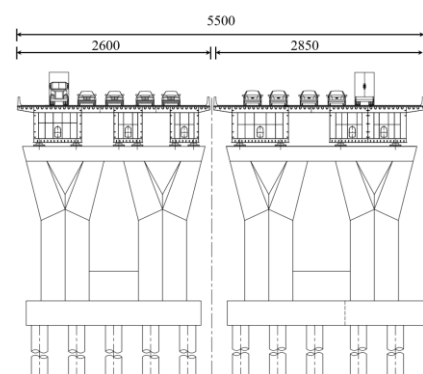


Figure 3. Cross-section of the new bridge (cm).

To ensure partial bridge operation during the reconstruction and expansion process, the construction conditions were designed to reflect the simultaneous demolition and reconstruction activities during the construction phase, as well as the conditions of the fully completed bridge. Considering the impact of upstream and downstream wind fields on the construction process, the construction conditions were established with the left carriageway designated as upstream and the right carriageway as downstream, as depicted in Figure 4. The specific construction conditions are detailed in Table 1, while the layout diagrams corresponding to each construction stage are presented in Figure 4.

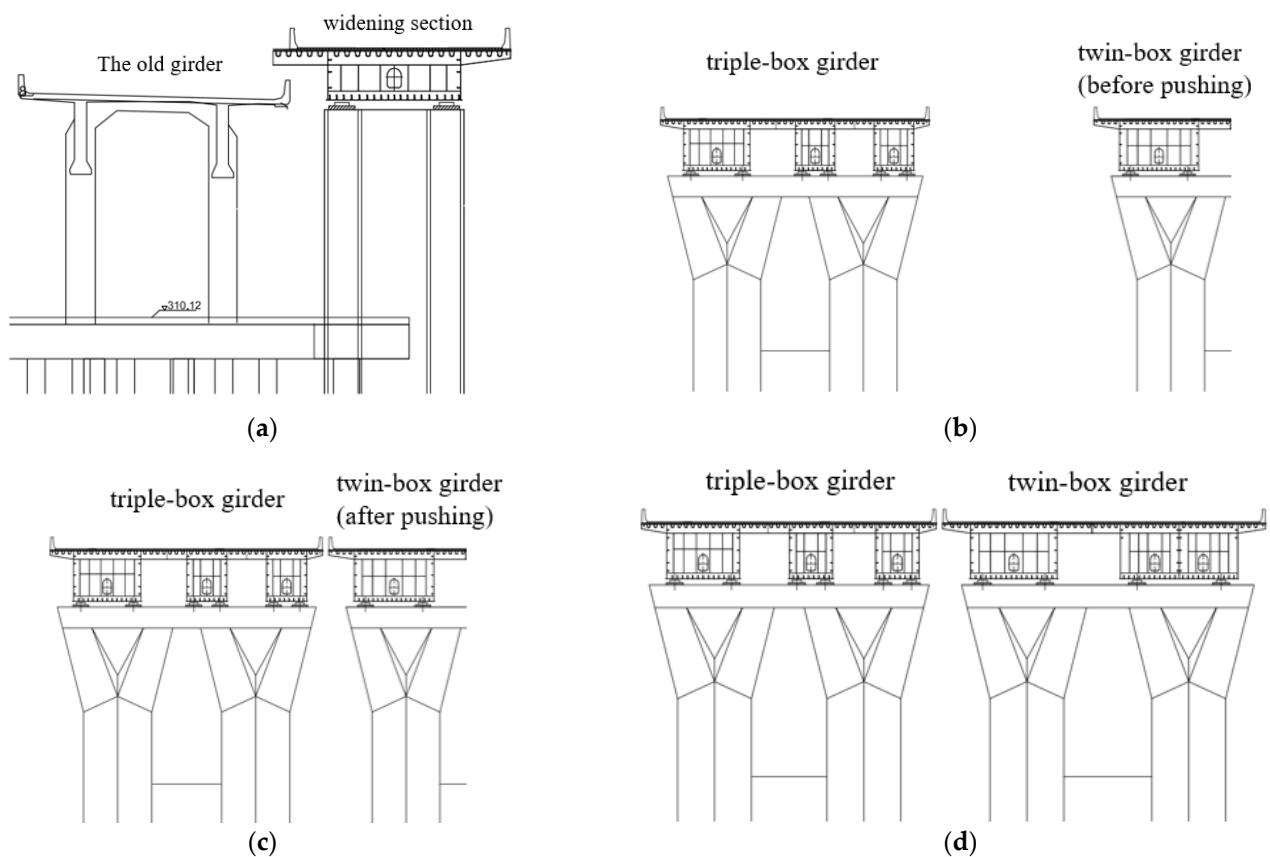


Figure 4. Layout diagrams for each construction stage. (a) Old Bridge and Widened Section; (b) New Bridge with Three-Box Main Girder and Widened Section; (c) New Bridge with Three-Box Main Girder and Widened Section; (d) New Bridge with Three-Box Main Girder and Two-Box Main Girder.

Table 1. Construction condition details.

Case Number	Construction Condition	Wind Attack Angle
DZ-01	widening section + the old bridge	$0^\circ, \pm 3^\circ$
DZ-02	the old bridge + widening section	$0^\circ, \pm 3^\circ$
DZ-03	triple-box girder of the new bridge + widening section	$0^\circ, \pm 3^\circ$
DZ-04	widening section + triple-box girder of the new bridge	$0^\circ, \pm 3^\circ$
DZ-05	triple-box girder of the new bridge + widening section (after pushing)	$0^\circ, \pm 3^\circ$
DZ-06	widening section (after pushing) + triple-box girder of the new bridge	$0^\circ, \pm 3^\circ$
DZ-07	twin-box girder of the new bridge + triple-box girder of the new bridge	$0^\circ, \pm 3^\circ$
DZ-08	triple-box girder of the new bridge + twin-box girder of the new bridge	$0^\circ, \pm 3^\circ$

3. CFD Simulation

To preliminarily determine the most unfavorable conditions during the construction phase, CFD simulation was conducted to analyze the bridge under construction. This simulation also serves as a preparatory step for subsequent wind tunnel experiments.

3.1. CFD Model Setup

The CFD simulation was performed using Fluent 2024R1 as the computational software. The $k-\omega$ turbulence model, which provides reliable results for external flow around bluff bodies, was selected, and steady-state flow calculations were employed. The SIMPLEC algorithm was used to solve the equations, with a time step of 0.001 s and a residual convergence criterion of 10^{-5} [31]. The computational model was established based on actual

bridge dimensions. The computational domain utilized triangular meshes, which adapt well to complex shapes. Denser mesh nodes were applied around the girder cross-section, with a gradual transition outward to coarser meshes away from the girder. To enhance computational accuracy, an additional refined mesh region was introduced within a small wind field. The total number of mesh elements was approximately 80,000. Boundary conditions were set with a velocity inlet on the left side, a pressure outlet on the right side, and symmetric boundaries on the upper and lower sides. The computational model and mesh discretization are shown in Figures 5 and 6, where B denotes the model width. The wind attack angles used in the simulation matched those in the experiments, specifically $\pm 3^\circ$ and 0° .

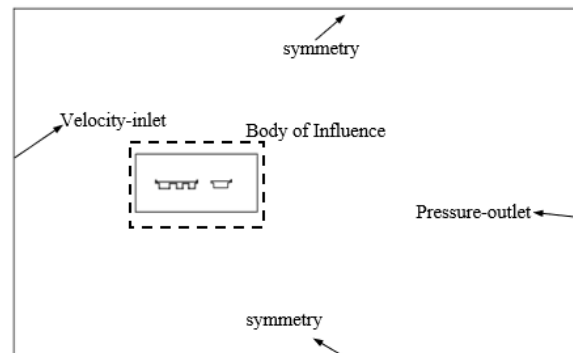


Figure 5. CFD computational domain division.

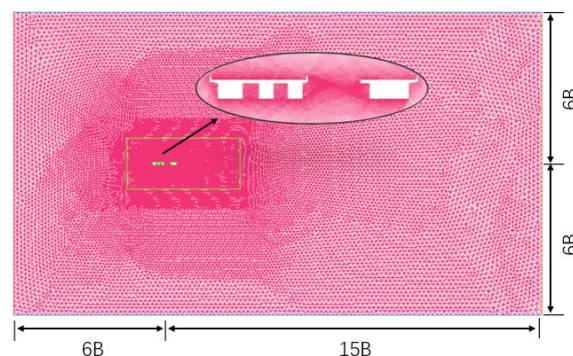


Figure 6. Mesh discretization diagram.

3.2. CFD Simulation Independence Test

CFD numerical simulations necessitate careful consideration of grid density's impact on computational accuracy. Optimal meshing requires not only sufficient grid resolution but also manageable computational time, rendering grid density selection a critical optimization process balancing numerical precision against time expenditure. The results are shown in Table 2. It can be seen from the table that the result is acceptable.

Table 2. Mesh independence verification.

Number of Cells	C_d	C_l	C_m
4.39×10^5	0.8295	−0.0192	0.0064
8.16×10^5	0.8201	−0.0172	0.0052
1.26×10^6	0.8215	−0.0173	0.0058

3.3. CFD Simulation Results

As illustrated in Figure 7, the flow field downstream of the upstream bridge develops organized vortex shedding patterns between parallel bridges, with wind attack angle variations significantly modulating this aerodynamic phenomenon. At 0° attack angle, vortices exhibit lower convective velocities while maintaining close adherence to structural surfaces. When angular deviation occurs, the flow acceleration zone (depicted in red) emerges above the structural surface, accompanied by increased vortex multiplicity in the inter-bridge region. Comparative analysis demonstrates that the complexity of vortex-induced flow patterns under 0° wind attack angle remains substantially reduced compared to other angular configurations, particularly in terms of flow separation coherence and three-dimensional interaction intensity.

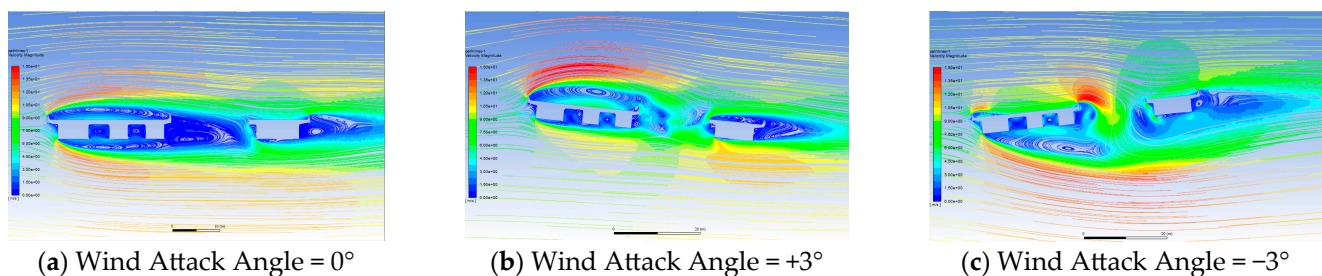


Figure 7. CFD Simulation of Wind Field Streamlines for the Three-Box Girder New Bridge with Widened Section.

Figure 8 illustrates the pressure distribution on the bridges under different wind attack angles. At a 0° wind attack angle, the upstream bridge experiences noticeably higher wind-induced pressure than the downstream bridge. However, changes in the wind attack angle alter the pressure contour between the upstream and downstream bridges, leading to the formation of pressure vortices. These vortices increase the pressure on the downstream bridge compared to the 0° condition, making the pressure distribution more complex and variable.

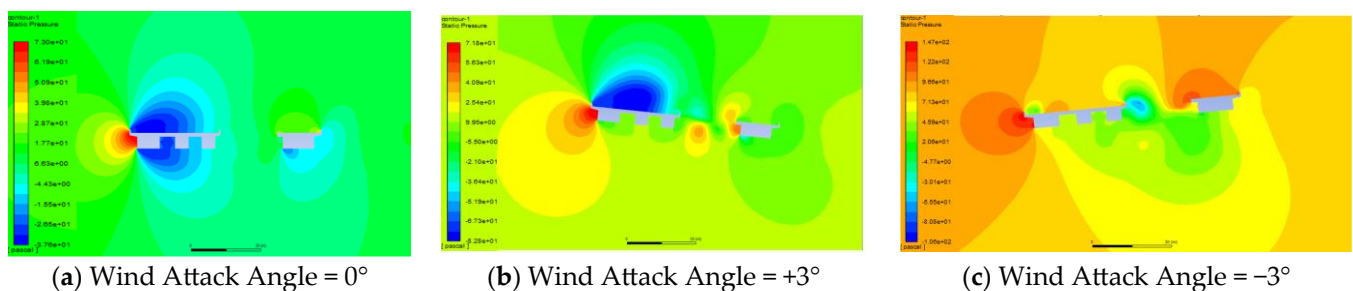


Figure 8. CFD Simulation of Wind Field Pressure Contours for the Three-Box Girder New Bridge with Widened Section.

3.4. Analysis of Vortex-Induced Vibration

The CFD results indicate that the flow field condition is the most complex under the dual carriageway partial-section construction scenario, where the maximum spacing between the newly constructed and demolished sections occurs simultaneously. The interference effects caused by the separation lead to significant upstream vortex shedding impacts on the single-box girder at the downstream position, making it prone to vibration effects induced by the upstream bridge's vortex shedding.

To approximate the VIV performance under these conditions, the Strouhal number for vortex shedding around a cylindrical shape was used to analyze the downstream section's single-box double-cell girder in the widened segment.

Investigation of wind-induced vibration phenomena revealed that, when fluid flows around a cylindrical structure, alternating vortices shed periodically in the wake. The vortex shedding frequency (f), wind velocity (v), and cylinder diameter (d) exhibit the following relationship:

$$St = \frac{fd}{v}$$

where St , Strouhal number (≈ 0.2 for circular cylinders); f , Vortex shedding frequency (Hz); d , Characteristic length (cylinder diameter); v , Oncoming flow velocity (m/s).

This formulation remains valid for bluff body sections, with d representing the characteristic dimension perpendicular to the flow. Through transient simulations in ANSYS Fluent 2024R1, the vortex shedding frequency f , can be numerically determined by monitoring pressure fluctuation spectra in the wake region.

Vortex-induced resonance starting wind speed of the structure at the first order vibration frequency:

$$v = \frac{f_v d}{St} = \frac{1.075 \times 5.5}{0.1726} = 34.26 \text{ m/s}$$

The calculations show that, at a structural fundamental vibration frequency, VIV resonance occurs when the wind speed reaches 34.26 m/s, triggering vortex-induced vibration. According to the preliminary estimates, the bridge exhibits a relatively low VIV onset wind speed, indicating a certain risk of VIV. These limitations underscore the imperative for comprehensive wind tunnel testing to rigorously validate and refine the evaluation of VIV characteristics under operational conditions.

4. Full-Bridge Aeroelastic Experiment

4.1. Overview of the Full-Bridge Aeroelastic Model

The full-bridge aeroelastic model test was conducted in the CA-03 atmospheric boundary layer wind tunnel at Chang'an University, with the primary objective of measuring the vibration amplitude of the bridge under the effects of mean and fluctuating winds. To ensure the reliability of the test results, the wind tunnel experiment adhered to the criteria of geometric similarity, dynamic similarity, and motion similarity. In general, aeroelastic similarity encompasses the structural similarity in terms of length, density, elasticity, and internal friction. Furthermore, parameters such as viscous forces, gravity, wind pressure, elasticity, mass, and damping in the wind tunnel test were required to align with those of the prototype.

The comparison between the target and measured natural frequencies of the aeroelastic model under various test conditions simulated in the wind tunnel is shown in Table 3, which also provides the damping ratios corresponding to the structural modes. The natural frequency data in Table 4 indicate that the relative error between the measured and target values of the first vertical bending frequency does not exceed $\pm 5\%$, demonstrating that the aeroelastic model exhibits high testing accuracy and satisfies the requirements of the wind tunnel experiment.

Table 3. Key parameters of the full-bridge aeroelastic model.

Parameter	Real Bridge Value		Similarity Ratio	Model Requirement Value
Length (m)		415	$\lambda_L = 1:50$	8.3
Width (m)	Main beam	Triple-box girder 26	$\lambda_B = 1:50$	Triple-box girder 0.52
		Twin-box girder 28.5	$\lambda_B = 1:50$	Twin-box girder 0.57
Height (m)	Main beam	3–5	$\lambda_H = 1:50$	0.06–0.1
Mass of unit length (kg/m)	Main beam	Triple-box girder 14,854	$\lambda_m = 1:50^2$	Triple-box girder 5.942
		Twin-box girder 16,424		Twin-box girder 6.570

Table 4. Dynamic characteristics of the aeroelastic model for the completed bridge.

Mode Description	Actual Theoretical (Hz)	Model Requirement (Hz)	Model Actual (Hz)	Error (%)
First order positive symmetric vertical bending frequency of Triple-box girder	1.117	7.783	7.432	−4.51%
First order positive symmetric vertical bending frequency of Twin-box girder	1.075	7.598	7.324	−3.61%

Considering the dimensions of the CA-03 atmospheric boundary layer wind tunnel (16.0 m in long, 15.2 m in width, and 2.0 m in height) and the need to accurately replicate bridge details, a geometric scale ratio of 1:50 was employed for the full-bridge aeroelastic model. Except for the Reynolds number and Cauchy number, all other dimensionless parameters were strictly simulated in the test. Table 2 presents the similarity relationships for the primary parameters of the full-bridge aeroelastic model. Although the Cauchy number similarity could not be fully achieved under the given test conditions, refined design ensured that the bending stiffness (EI) and torsional stiffness (GJ) satisfied the required similarity ratios. This approach guarantees the credibility and scientific rigor of the experimental results.

In the aeroelastic model of the bridge structure, the rotational stiffness is entirely provided by the model's framework, with aluminum selected as the core girder material. The material selection for the scaled model prioritized strict compliance with mass and frequency similarity requirements. Initial steel implementation resulted in excessive mass exceeding prescribed design parameters. To resolve this dynamic scaling conflict, aluminum alloy was ultimately employed as the girder material, achieving optimal balance between structural mass distribution and fundamental vibration characteristics while maintaining requisite stiffness properties. For the piers, core girders were designed to meet the similarity requirements for transverse and longitudinal bending stiffness. The steel framework's axis was precisely aligned with the actual axis of the bridge.

To meet the geometric similarity ratio requirements, the external appearance of the model was constructed using lightweight wooden panels, which were manually carved, adhered, and bolted to the core girder to form an integrated structure. This configuration effectively replicated the external features of the actual bridge. Figure 9 illustrates the full-bridge aeroelastic model of the structure.



Figure 9. Full-bridge aeroelastic model.

4.2. Verification of Structural Dynamic Characteristics

After the installation and calibration of the aeroelastic model, structural dynamic characteristic testing was conducted to ensure that the dynamic properties of the model's main girder met the requirements of the wind tunnel experiment.

The modal testing was performed using the free decay excitation method under zero wind speed conditions. The modal testing system consisted of signal sensors, data acquisition equipment, a PC, and analysis software. The signal sensor employed was the HL-G112-S-J laser displacement sensor manufactured by Panasonic, Tokyo, Japan, to measure the vibration displacement of the model. The data acquisition system utilized the DASPV11 data acquisition analyzer developed by the Oriental Institute of Vibration and Noise Technology, Beijing, China, integrated with a PC. The installation layout of the experimental equipment is illustrated in Figure 10.

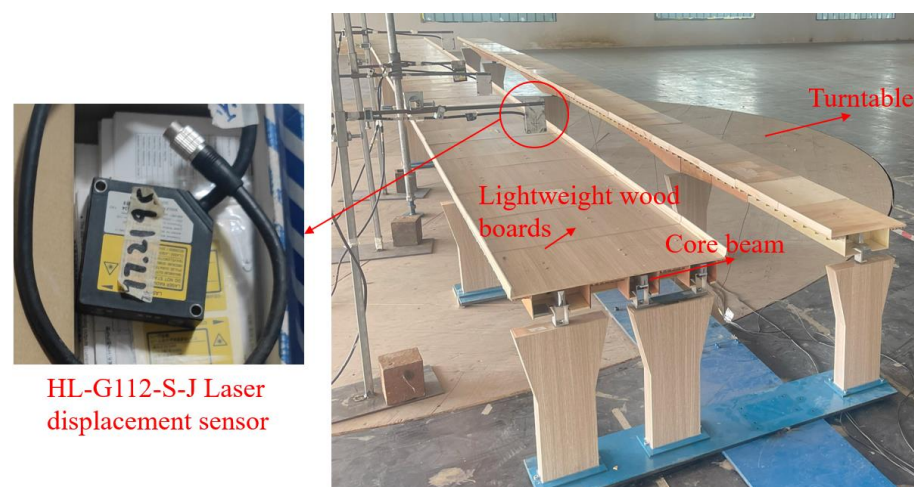


Figure 10. Full-bridge aeroelastic experiment setup.

For the primary test conditions of the bridge structure in its completed stage, the free vibration method was used to test the natural vibration characteristics, including the first vertical bending mode of the aeroelastic model. By analyzing the displacement signals obtained from dynamic characteristic testing, the damping ratios of the structural modes were evaluated to verify whether the damping performance of the aeroelastic model met the experimental requirements.

Given the symmetric structure of the new bridge, the selection of measurement points adhered to the principles of representativeness and uniform distribution [32]. Ultimately, the mid-span half-point and quarter-point of the main span, as well as the half-point of a side span, were chosen as vibration measurement locations to obtain critical data.

4.3. Experimental Results and Analysis

Based on the time–history data collected by laser displacement sensors, curves depicting the variation in bridge vibration amplitude with wind speed under different scenarios were plotted. Figure 11 illustrates the bridge responses under various construction stages and wind attack angles. The vibration amplitude limit for all scenarios was set at 61.5 mm. The figure shows that no large-amplitude vibrations occurred under any scenario, with a maximum recorded amplitude of 24.59 mm, indicating that the bridge satisfies the requirements of JTG/T 3360-01-2018 Wind-Resistant Design Specifications for Highway Bridges [33].

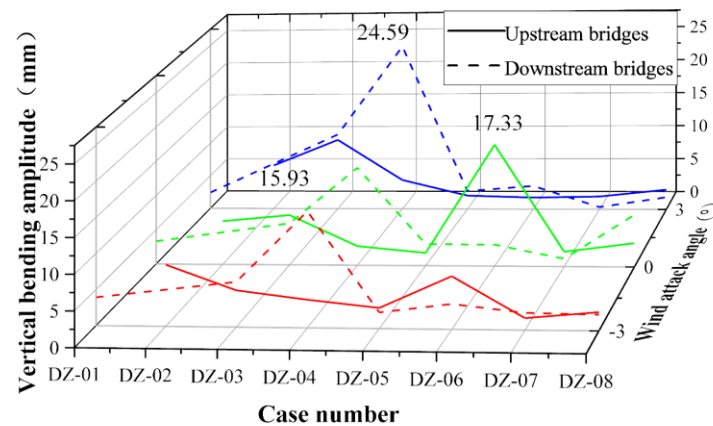


Figure 11. Maximum Amplitude for Each Scenario.

Further analysis reveals that the scenarios with relatively large amplitudes are DZ-04 and DZ-06, both featuring configurations where the widened section is located upstream in the wind field, and the three-box girder is located downstream. However, their vibration characteristics differ: when the widened section is farther from the three-box girder, the amplitude of the three-box girder increases significantly; when the widened section is closer to the three-box girder, the widened section experiences increased vibration amplitudes.

As shown in Figure 12, the three-box girder exhibited the maximum vibration amplitude during the DZ-04 scenario, with a peak amplitude of 24.59 mm at a wind attack angle of 3°. At this angle, the amplitude reached its maximum at a real bridge wind speed of 35.36 m/s before gradually decreasing, followed by another increase at 43.84 m/s. When the wind attack angle was 0°, the amplitude peaked at a real bridge wind speed of 38.18 m/s before decreasing, then increased again at 46.67 m/s. At a wind attack angle of −3°, the onset wind speed for vibrations was similar to that at 0°.

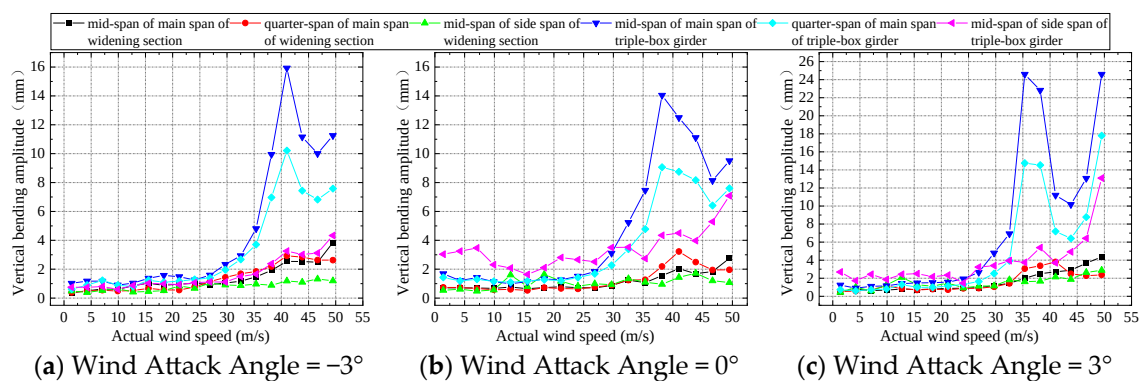


Figure 12. Bridge amplitude under DZ-04 scenario at different wind attack angles and at different wind speeds.

The analysis indicates that the three-box girder is affected by the wind field generated by the upstream widened section, resulting in VIV, with the maximum amplitude occurring at wind speeds between 35 and 38 m/s. These results align with the CFD-based estimates of the critical wind speeds for VIV (35 m/s and 50 m/s), confirming that vortices generated by the upstream bridge significantly influence the downstream bridge. However, this impact is limited to specific wind speeds, and the effect on vibration amplitude is negligible when the actual wind speed deviates from this range.

As shown in Figure 13, another scenario with significant vibration amplitude, DZ-06, was selected for analysis. Under this scenario, the maximum bridge vibration amplitude reached 17.33 mm, accounting for 70.48% of the maximum amplitude observed in the DZ-04 scenario. However, a key difference is that the maximum amplitude for DZ-06 occurred at a wind attack angle of 0° , while no significant vibration was observed at a 3° wind attack angle. In this scenario, the real bridge wind speed corresponding to the larger vibration amplitude was consistently 38.18 m/s. As the real bridge wind speed increased further, the vibration amplitude showed a significant reduction. The real bridge wind speed at which VIV occurred had an error of less than 10% compared to the previously calculated value of 35 m/s in the preceding sections of this study.

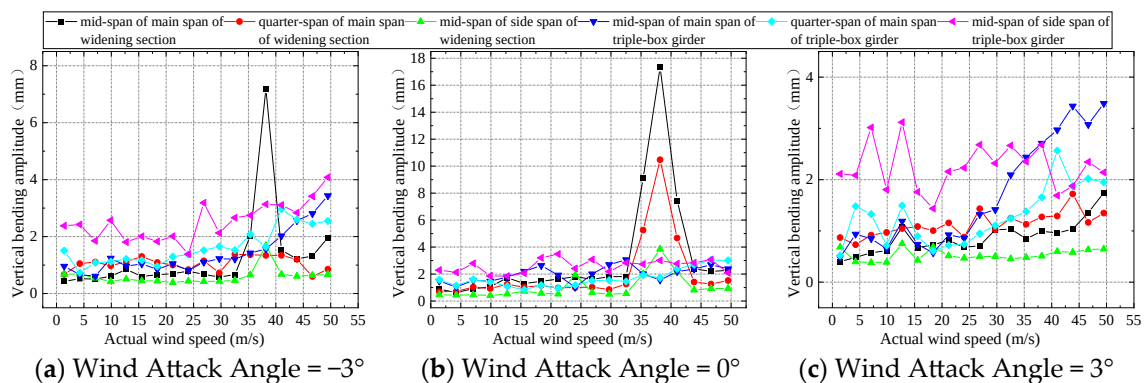


Figure 13. Bridge amplitude under DZ-06 scenario at different wind attack angles and at different wind speeds.

5. Conclusions

This study focuses on potential VIV issues arising during the simultaneous demolition of old continuous girder bridges and the construction of new ones. Using a combination of CFD numerical simulations and wind tunnel experiments, the aerodynamic performance and VIV characteristics of the bridge during different construction stages were systematically analyzed. The research includes a comparative analysis of CFD simulations and wind tunnel results, the impact of girder spacing on VIV during construction, and a discussion on VIV risk assessment and mitigation strategies. The following conclusions were drawn:

- (1) Through an in-depth comparison of CFD simulation results and wind tunnel experiments, the effectiveness of using the Strouhal number for cylindrical vortex shedding as an approximate analysis method was validated. This approach plays a significant role in guiding the demolition of old bridges and the simultaneous construction of new ones. It enables the rapid estimation of the critical wind speed for VIV, providing valuable preliminary predictions of wind-resistant performance during the early stages of bridge design. Additionally, this method offers direction for wind tunnel testing, allowing the focus to be placed on critical wind speed ranges, thereby reducing costs and improving efficiency. It provides essential wind-induced safety assurance for subsequent construction phases.

- (2) A critical phenomenon observed in this experimental study revealed that, when the spacing between the three-box girder and half-width girder configurations reached its maximum operational condition (specifically one times the girder width of half-width girder), the downstream bridge exhibited a peak VIV amplitude of 24.59 mm—the highest recorded value in this test series. Unlike conventional VIV characteristics, this particular oscillation demonstrated a “jumping” pattern without a distinct lock-in region. Complementary CFD simulations identified intensified flow field complexity between the upstream and downstream bridge sections. These findings emphasize the necessity in parallel new-existing bridge construction planning to maintain an inter-bridge spacing within one girder width, thereby mitigating flow field interference effects and ensuring structural safety of downstream bridges. Consequently, future bridge design and construction strategies should prioritize effective management and optimization of inter-bridge spatial relationships to alleviate such aerodynamic interactions.
- (3) Based on the detailed results from the CFD simulations and wind tunnel experiments, there is a certain risk of VIV during the renovation process of old and new bridges, particularly under specific wind speed conditions. Therefore, potential VIV issues must be fully considered during the early design stages. CFD simulations can provide preliminary assessments of bridge VIV risks, guiding further verification through wind tunnel experiments. Additionally, the wind environment characteristics of the construction site should be analyzed in depth. For potential VIV problems, early optimization of aerodynamic performance should be considered, or temporary wind-resistant measures such as adjustable dampers or temporary wind barriers can be implemented during construction. These measures will significantly enhance the wind resistance of the old and new bridges at different construction stages, reduce potential risks from VIV, and provide scientific guidance for bridge renovation and design.

In conclusion, this study systematically explores the prediction and mitigation of VIV issues, providing valuable reference data for safe bridge construction and wind-resistant design.

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Conflicts of Interest: Authors Fulin Yang and Xinmin Zhang were employed by the company Guangshen Expressway Reconstruction and Expansion Management Office. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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